3 Fig. 3.1 shows an archer aiming an arrow at a target.

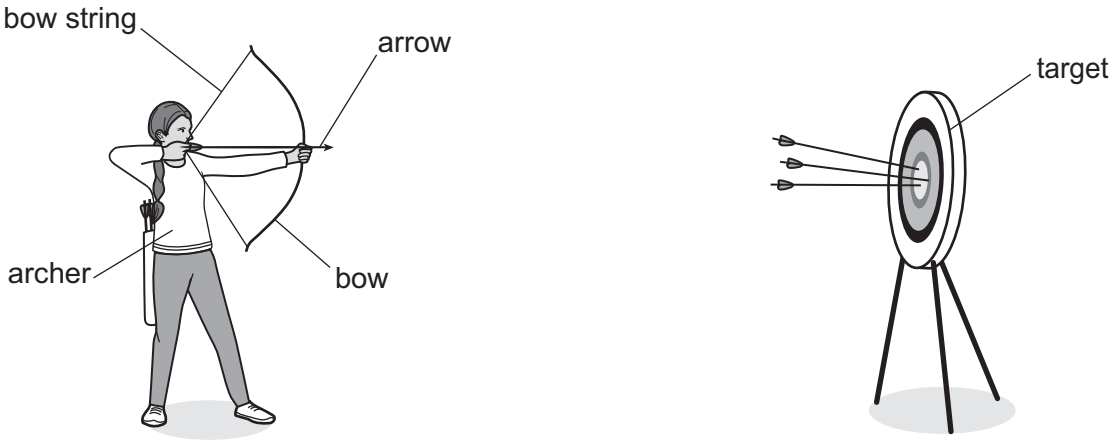

Fig. 3.1

(a) The archer pulls back on the bow string, doing a total of 110 J of work. Her hand moves a distance of 0.45 m. The bow is bent and stores energy.

Show that the average force applied by the archer in pulling the string back is approximately 240 N.

[2]

(b) The archer releases the bow string. All the energy in **(a)** is transferred to the arrow.

The arrow moves off at an initial speed *v*. The mass of the arrow is 0.030 kg.

(i) Calculate the initial speed *v* of the arrow.

initial speed *v* = ... [3]

(ii) Explain why the speed of the arrow as it hits the target is less than the value in **(b)(i)**.

.....

..... [2]

[Total: 7]

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